Cycle-lap rigidity for synchronized negative shadows

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Let

$$2^{a_j} x_{j+1} = 3x_j + 1, \qquad 2^{a_j} e_{j+1} = 3e_j - 1, \qquad a_j \geq 1, \tag{1}$$

where the x_j are positive odd integers and the e_j are real numbers greater than 1. Put

$$m_j := \lceil e_j \rceil, \qquad \theta_j := m_j - e_j \in [0, 1). \tag{2}$$

Suppose that, on an interval of indices, there are a fixed $c \in \{1, 2, 3\}$ and positive odd integers y_j such that

$$m_j = cy_j, \qquad 2^{a_j} y_{j+1} = 3y_j - 1. \tag{3}$$

Thus (y_j) is an accelerated positive $3y - 1$ shadow synchronized with the positive $3x + 1$ path.

Theorem 1 (synchronized cycle-lap rigidity)**.** *Assume that (3) holds from p through $p + KL$, where $K, L \geq 1$, and that*

$$y_{p+L} = y_p. \tag{4}$$

Put

$$M := y_p, \qquad B := \sum_{i=0}^{L-1} a_{p+i}, \qquad \rho := \frac{3^L}{2^B}. \tag{5}$$

Then the shadow and exponent blocks of length L repeat throughout the displayed interval, and

$$\boxed{\begin{aligned} &\rho > 1, \\ &v_2(x_p + M) = KB + v_2(x_{p+KL} + M) \geq KB + 1, \\ &x_{p+KL} + M = \rho^K (x_p + M). \end{aligned}} \tag{6}$$

Moreover,

$$\boxed{\log \rho = \sum_{i=0}^{L-1} -\log\left(1 - \frac{1}{3y_{p+i}}\right), \qquad \sum_{i=0}^{KL-1} \frac{1}{m_{p+i}} < \frac{3}{c} \log \frac{x_{p+KL} + M}{x_p + M}.} \tag{7}$$

The rounding error at the first return is exponentially pinched. More precisely,

$$\boxed{\begin{aligned} c = 1 \quad &\Longrightarrow \quad 0 \le \theta_p < \rho^{-K}, \\ c = 2,\ M = 1 \quad &\Longrightarrow \quad \rho = \frac{3}{2}, \qquad 0 < 1 - \theta_p \le \left(\frac{2}{3}\right)^K, \\ (c-1)M > 1 \quad &\Longrightarrow \quad \rho^K < \frac{(c-1)M}{(c-1)M - 1}. \end{aligned}} \tag{8}$$

If p is chosen at a maximum state M of the shadow cycle, then

$$\rho \ge 1 + \frac{1}{2M}. \tag{9}$$

Consequently,

$$\boxed{c = 3 \Longrightarrow K \le 1, \qquad c = 2,\ M \ge 3 \Longrightarrow K \le 2.} \tag{10}$$

Thus arbitrarily many complete laps are possible only for a color-1 shadow cycle or for the color-2 fixed shadow $y = 1$.

Proof. Because y_{j+1} is odd, the second identity in (3) determines both

$$a_j = v_2(3y_j - 1) \qquad \text{and} \qquad y_{j+1} = \frac{3y_j - 1}{2^{v_2(3y_j - 1)}}. \tag{11}$$

The shadow map is deterministic. Equation (4) therefore makes both the shadow states and the exponents L-periodic for as long as (3) holds.

Let C be the positive affine carry obtained by iterating one period of the negative recurrence. We have

$$2^B M = 3^L M - C, \qquad C > 0. \tag{12}$$

Hence $(3^L - 2^B)M = C$, which proves $\rho > 1$. Adding the two synchronized recurrences gives

$$2^{a_j}(x_{j+1} + y_{j+1}) = 3(x_j + y_j). \tag{13}$$

After K periods,

$$2^{KB}(x_{p+KL}+M) = 3^{KL}(x_p+M). \tag{14}$$

The two powers in (14) are coprime. Since x_{p+KL} and M are odd, taking 2-adic valuations gives

$$v_2(x_p+M) = KB + v_2(x_{p+KL}+M) \geq KB+1.$$

Division by 2^{KB} gives the last identity in (6).

For one shadow edge we have

$$\frac{y_{j+1}}{y_j} = \frac{3}{2^{a_j}}\left(1-\frac{1}{3y_j}\right). \tag{15}$$

The product over one cycle equals 1. Taking logarithms proves the first identity in (7). Since $-\log(1-t) > t$ for $0 < t < 1$, it follows that

$$\sum_{i=0}^{L-1} \frac{1}{y_{p+i}} < 3\log\rho. \tag{16}$$

The same L states occur in every lap and $m_j = cy_j$. Thus

$$\sum_{i=0}^{KL-1} \frac{1}{m_{p+i}} = \frac{K}{c}\sum_{i=0}^{L-1}\frac{1}{y_{p+i}} < \frac{3K}{c}\log\rho. \tag{17}$$

Equation (6) turns the last member into the logarithmic ratio in (7).

Put $\Delta_j := e_j - y_j$. Subtracting the two negative recurrences gives

$$2^{a_j}\Delta_{j+1} = 3\Delta_j, \qquad \Delta_{p+KL} = \rho^K\Delta_p. \tag{18}$$

At a return to the state M, equations (2) and (3) give

$$(c-1)M - 1 < \Delta_j = (c-1)M - \theta_j \leq (c-1)M. \tag{19}$$

If $c = 1$, equation (18) says $\theta_{p+KL} = \rho^K\theta_p < 1$, which proves the first line of (8). If $c = 2$ and $M = 1$, then the only positive shadow cycle is the fixed edge $1 \mapsto 1$, with exponent 1 and multiplier 3/2. Now (18)–(19) give

$$0 < 1-\theta_p = \rho^{-K}(1-\theta_{p+KL}) \leq \rho^{-K}. \tag{20}$$

Finally, if $(c-1)M > 1$, then (18)–(19) give

$$\rho^K\big((c-1)M-1\big) < (c-1)M, \tag{21}$$

which is the last line of (8).

We now rotate the cycle so that its displayed initial state M is maximal. If a is the exponent on the edge returning to M, then

$$2^a M = 3y - 1 \le 3M - 1, \tag{22}$$

where $y \le M$ is the preceding cycle state. Since $a \ge 1$, equation (22) forces $a = 1$. Writing the carry in (12) as

$$C = \sum_{i=0}^{L-1} 3^{L-1-i} 2^{B_i}, \qquad B_i := \sum_{h=0}^{i-1} a_{p+h}, \tag{23}$$

we see that its last normalized term is $2^{B_{L-1}-B} = 2^{-a} = 1/2$. Therefore

$$(\rho - 1)M = \frac{C}{2^B} \ge \frac{1}{2}, \tag{24}$$

which proves (9).

Bernoulli's inequality and (9) give

$$\rho^K \ge 1 + K(\rho - 1) \ge 1 + \frac{K}{2M}. \tag{25}$$

For $c = 3$, equations (8) and (25) imply

$$K < \frac{2M}{2M - 1} < 2,$$

so $K \le 1$. For $c = 2$ and $M \ge 3$, they imply

$$K < \frac{2M}{M - 1} \le 3,$$

so $K \le 2$. This proves (10). □

Corollary 2 (the two unbounded-lap mechanisms)**.** *Consider a maximal constant nonpositive-defect run from the synchronized ceiling construction, and write* $c := 1 - d \in \{1, 2, 3\}$*. Use its positive odd shadow*

c	*shadow states*
1	$y_j = m_j$,
2	$y_j = m_j/2$ *apart from the last ceiling state,*
3	$y_j = m_j/3$ *apart from the first ceiling state.*

(26)

Let $H := 10.789$ *be the certified bound for the reciprocal mass of every finite simple positive accelerated* $3y - 1$ *path. Then the following hold.*

1. *A color-3 shadow traverses any one cycle at most once. Its whole ceiling run has reciprocal mass less than*

$$H + \frac{1}{3}. \tag{27}$$

2. *In a color-2 shadow, every cycle other than the fixed point* 1 *is traversed at most twice. After deleting the terminal block on which* $m_j = 2$*, the reciprocal mass of the ceiling run is less than*

$$2H + \frac{1}{2}. \tag{28}$$

If that terminal block contains K *edges and begins at* p*, then all of its exponents equal* 1 *and*

$$\boxed{\begin{aligned} &2^{K+1} \mid x_p + 1, \\ &x_{p+K} + 1 = \left(\frac{3}{2}\right)^K (x_p + 1), \\ &0 < 1 - \theta_p \le \left(\frac{2}{3}\right)^K. \end{aligned}} \tag{29}$$

3. *In a color-1 run which contains two occurrences of a maximum state* M *on its eventual shadow cycle, choose the first such occurrence* p *and the last one* $q = p + KL$*. The indices outside the complete laps* $[p, q)$ *form at most two simple shadow pieces. Hence*

$$\boxed{\begin{aligned} &\sum_{j \text{ in the run}} \frac{1}{m_j} < 2H + 3 \log \frac{x_q + M}{x_p + M}, \\ &2^{KB+1} \mid x_p + M, \\ &\theta_p < \rho^{-K}. \end{aligned}} \tag{30}$$

If no cycle maximum occurs twice, the run is covered by at most two simple shadow pieces and has mass less than $2H$*. If the complete shadow is simple, the sharper bound* H *holds.*

*Consequently, unbounded reciprocal mass inside individual nonpositive defect runs can only be supplied by color-*1 *cycle laps or by terminal color-*2 *towers with* $m = 2$ *and exponent* 1*.*

Proof. A finite path under the deterministic accelerated $3y-1$ map is simple until its first repeated state. After that repetition it follows one cycle. Every transient piece, every single traversal of the cycle with its repeated endpoint removed, and every proper terminal piece is simple, so each has reciprocal mass less than H.

For $c = 3$, the theorem allows at most one complete cycle lap. We split the shadow into a simple initial piece, that lap, and a simple terminal piece. Their total shadow mass is less than $3H$. Division by $c = 3$ and inclusion of the discarded initial ceiling state, whose value is at least 3, prove (27).

For $c = 2$, a nontrivial cycle has maximum at least 3 and is traversed at most twice. Away from the fixed point 1, the shadow is therefore covered by a simple initial piece, at most two simple cycle laps, and a simple terminal piece. Its shadow mass is less than $4H$. Division by 2 and inclusion of the discarded last ceiling state prove (28). Once the shadow reaches 1, determinism gives the fixed edge $1 \mapsto 1$ with exponent 1. Formula (29) is (6) and (8) for this fixed cycle.

For $c = 1$, the shadow is the complete ceiling run. If a cycle maximum occurs at least twice, the portion before its first chosen occurrence and the portion beginning at its last chosen occurrence are simple. Their combined mass is less than $2H$. Apply (6)–(8) to the complete laps between them to obtain (30). If no cycle maximum occurs twice, splitting at the first repeated state covers the path by at most two simple pieces. The completely simple case is the definition of H. □

Theorem 3 (fixed-seed plateau shell for the exceptional towers)**.** *Assume now that (1) is the fixed synchronized companion of a normalized divergent orbit. Put*

$$A_j := \sum_{i<j} a_i, \qquad q_j := \frac{3^j}{2^{A_j}}, \tag{31}$$

and let $\mathcal{K}$ be its stationary constant, so that

$$x_j + e_j = q_j \mathcal{K}. \tag{32}$$

Define the centered rational anchor and its reduced ternary depth by

$$\Xi_j := \frac{x_j + 1}{q_j} = \frac{2^{A_j}(x_j + 1)}{3^j}, \qquad D_j := \max\{0, j - v_3(x_j + 1)\}. \tag{33}$$

Then

$$\boxed{\Xi_{j+1} - \Xi_j = \frac{2^{a_j} - 2}{3q_j} \geq 0, \qquad \mathcal{K} - \Xi_j = \frac{e_j - 1}{q_j} > 0.} \tag{34}$$

In particular, Ξ_j is nondecreasing and changes exactly when $a_j > 1$.

Let $[p, p+K]$ be an $m = 2$ tower of $K \geq 1$ edges, meaning that

$$m_{p+i} = 2, \qquad a_{p+i} = 1 \qquad (0 \leq i < K), \tag{35}$$

and that $m_{p+K} = 2$. Put

$$b := v_2(x_p + 1), \qquad v := v_3(x_p + 1), \qquad h := \frac{x_p + 1}{2^b 3^v}. \tag{36}$$

Then $b \geq K + 1$, $(h, 6) = 1$, and, for $0 \leq i \leq K$,

$$\boxed{x_{p+i} + 1 = 2^{b-i} 3^{v+i} h, \qquad \Xi_{p+i} = \Xi_p, \qquad D_{p+i} = D_p.} \tag{37}$$

The reduced denominator of Ξ_p is exactly 3^{D_p}.

Distinct maximal $m = 2$ towers have distinct, strictly increasing anchors. Consequently, for every integer $D \geq 0$, the complete fixed orbit contains at most

$$\boxed{\lceil \mathcal{K} 3^D \rceil - 1} \tag{38}$$

maximal towers whose reduced ternary depth is at most D.

Consider two consecutive maximal towers. Let their anchors be $\xi_i < \xi_{i+1}$ and their depths be D_i, D_{i+1}. Let r_i be the state index at which the first tower ends. Then

$$\boxed{3^{-\max\{D_i, D_{i+1}\}} \leq \xi_{i+1} - \xi_i < \mathcal{K} - \xi_i \leq \frac{1}{q_{r_i}}.} \tag{39}$$

Hence

$$\boxed{\max\{D_i, D_{i+1}\} > \log_3 q_{r_i}.} \tag{40}$$

Finally, a tower beginning at p satisfies the two fixed-seed length bounds

$$\boxed{K + 1 + v_3(x_p + 1) \log_2 3 < \log_2 q_p + \log_2 \mathcal{K},} \tag{41}$$

and, with $n := x_0$,

$$\boxed{K + 1 + v_3(x_p + 1) \log_2 3 \leq p \log_2 \frac{3}{2} + \log_2(n + 1).} \tag{42}$$

Thus an infinite family of exceptional towers must escape every bounded reduced ternary shell, and every adjacent pair must expose a shell at least as deep as the current coefficient surplus measured in base 3.

Proof. Put $X_j := x_j + 1$. The positive recurrence in (1) gives

$$2^{a_j} X_{j+1} = 3X_j + 2^{a_j} - 2. \tag{43}$$

After division by $2^{a_j} q_{j+1} = 3q_j$, this becomes the first identity in (34). The second follows by subtracting $X_j = x_j + 1$ from (32).

On the tower, equation (43) reduces to

$$X_{j+1} = \frac{3}{2} X_j. \tag{44}$$

Iteration proves the first identity in (37). Since X_{p+K} is even, its 2-adic valuation $b - K$ is at least 1, so $b \geq K + 1$. The definition of h gives $(h, 6) = 1$. Along the tower we also have

$$A_{p+i} = A_p + i, \qquad v_2(X_{p+i}) = b - i, \qquad v_3(X_{p+i}) = v + i. \tag{45}$$

Substitution in (33) proves the other two identities in (37). Cancellation of the exact power $3^{\min\{p,v\}}$ in (33) shows that the remaining denominator is precisely 3^{D_p}.

Suppose that two maximal towers occur in this order. If every intervening exponent were 1, then (1) would give

$$e_{j+1} - 1 = \frac{3}{2}(e_j - 1) \tag{46}$$

throughout the intervening interval. Once this increasing sequence leaves the ceiling cell $1 < e_j \leq 2$, it cannot return. Hence some intervening exponent is greater than 1. Equation (34) proves that the second anchor is strictly larger than the first.

If $D_j \leq D$, then the reduced anchor can be written as a positive integer multiple of 3^{-D}. All anchors lie strictly between 0 and $\mathcal{K}$ by (34). Their strict increase therefore permits at most $\lceil \mathcal{K} 3^D \rceil - 1$ such values. This proves (38).

Two distinct rationals with reduced denominators 3^{D_i} and $3^{D_{i+1}}$ differ by at least $3^{-\max\{D_i, D_{i+1}\}}$. At the endpoint r_i of the first tower, its anchor is still ξ_i, while $m_{r_i} = 2$ gives

$$0 < e_{r_i} - 1 \leq 1. \tag{47}$$

The second identity in (34) now proves every inequality in (39). Taking reciprocals of the two outer bounds gives (40).

The coprime divisors in (36) and the inequality $b \geq K+1$ give

$$2^{K+1}3^{v} \leq X_p. \tag{48}$$

On the other hand, (34) gives $X_p < q_p\mathcal{K}$. Taking binary logarithms proves (41).

It remains to give a seed-only upper bound. From (43), for every $a_j \geq 1$ and $X_j \geq 2$ we have

$$\begin{aligned} X_{j+1} &= \frac{3X_j + 2^{a_j} - 2}{2^{a_j}} \\ &\leq \frac{3}{2}X_j, \end{aligned} \tag{49}$$

because equality holds when $a_j = 1$, while for $a_j \geq 2$ the difference after clearing denominators is $(2^{a_j-1}-1)(3X_j-2) \geq 0$. Thus

$$X_p \leq \left(\frac{3}{2}\right)^{p}(n+1). \tag{50}$$

Combining (48) and (50), then taking binary logarithms, proves (42). □

Theorem 4 (fixed-seed plateau shell for zero-defect laps)**.** *Retain the normalized fixed-seed setup of the preceding theorem. Put*

$$N_j := x_j + m_j, \qquad d_j := 2^{a_j}m_{j+1} - 3m_j + 1, \tag{51}$$

and define the upper rational anchor and its reduced ternary depth by

$$\Omega_j := \frac{N_j}{q_j} = \frac{2^{A_j}N_j}{3^{j}}, \qquad E_j := \max\{0, j - v_3(N_j)\}. \tag{52}$$

Then

$$\boxed{\Omega_{j+1} - \Omega_j = \frac{d_j}{3q_j}, \qquad \Omega_j - \mathcal{K} = \frac{\theta_j}{q_j} \geq 0.} \tag{53}$$

Every maximal zero-defect interval is finite. On such an interval, Ω_j and E_j are constant, and the reduced denominator of its anchor is exactly 3^{E_j}.

For every integer $E \geq 0$, only finitely many maximal zero-defect intervals have reduced ternary depth at most E.

Let two maximal zero-defect intervals have distinct anchors ω_i, ω_{i+1} and depths E_i, E_{i+1}. For each interval, let

$$Q_i := \max\{q_j : j \text{ is a state index in that interval}\}. \tag{54}$$

Then

$$\boxed{3^{-\max\{E_i,E_{i+1}\}} \le |\omega_{i+1}-\omega_i| < \frac{1}{Q_i}+\frac{1}{Q_{i+1}} \le \frac{2}{\min\{Q_i,Q_{i+1}\}}.} \tag{55}$$

Consequently,

$$\boxed{\max\{E_i,E_{i+1}\} > \log_3 \frac{\min\{Q_i,Q_{i+1}\}}{2}.} \tag{56}$$

Suppose that one zero-defect interval contains $J \ge 1$ complete laps of a positive odd $3y-1$ cycle. Choose a maximum cycle state M, let the cycle have length L and total exponent B, and let p be the first selected occurrence of M. Then

$$\boxed{JB + 1 + v_3(x_p+M)\log_2 3 < \log_2 q_p + \log_2(\mathcal{K}+1).} \tag{57}$$

Thus repeated zero-defect laps also escape every bounded ternary shell, and their total exponent is paid by the coefficient surplus at their moving start.

Proof. Adding the positive recurrence in (1) to the ceiling recurrence obtained from (51) gives

$$2^{a_j} N_{j+1} = 3N_j + d_j. \tag{58}$$

Division by $2^{a_j} q_{j+1} = 3q_j$ proves the first identity in (53). The identities $N_j = x_j + m_j$ and $x_j + e_j = q_j\mathcal{K}$ prove the second.

When $d_j = 0$, equation (58) shows that $\Omega_{j+1} = \Omega_j$ and

$$v_3(N_{j+1}) = v_3(N_j) + 1. \tag{59}$$

Hence $E_{j+1} = E_j$. Cancellation in (52), exactly as in the proof of (37), shows that the reduced denominator is 3^{E_j}. A zero-defect tail is impossible: iteration of (58) would force its unbounded accumulated exponent to divide the fixed nonzero integer at the initial state. Thus every maximal zero-defect interval is finite.

We next prove finiteness in a bounded shell. Since the normalized orbit has $q_j > 1$ for $j \ge 1$ and $0 \le \theta_j < 1$, equation (53) places every upper anchor in the bounded interval

$$\mathcal{K} \le \Omega_j < \mathcal{K}+1. \tag{60}$$

There are only finitely many rationals in this interval whose reduced denominator divides 3^E. Fix one such rational R. If $R > \mathcal{K}$, then (53) gives

$$q_j < \frac{1}{R-\mathcal{K}} \tag{61}$$

at every occurrence of R. Since $q_j \to \infty$, this happens only finitely often.

It remains to consider $R = \mathcal{K}$. If $\mathcal{K}$ is irrational, this rational equality is impossible. If $\mathcal{K} = u/v$ in lowest terms, then an occurrence would make

$$q_j \mathcal{K} = \frac{u 3^j}{v 2^{A_j}} \tag{62}$$

an integer. The 2-adic valuation of the right side tends to $-\infty$ because $A_j \to \infty$. Thus (62) also has only finitely many solutions. This proves bounded-shell finiteness.

Two distinct anchors with reduced denominators 3^{E_i} and $3^{E_{i+1}}$ differ by at least the first member of (55). On the other hand, (53) and the definition of Q_i give

$$0 \le \omega_i - \mathcal{K} < \frac{1}{Q_i}, \qquad 0 \le \omega_{i+1} - \mathcal{K} < \frac{1}{Q_{i+1}}. \tag{63}$$

The triangle inequality proves the middle bound in (55), and the last bound is immediate. Rearrangement gives (56).

Finally, on a zero-defect cycle lap we have $m_j = y_j$. At the selected maximum state, $N_p = x_p + M$. The cycle-lap theorem gives

$$2^{JB+1} \mid N_p. \tag{64}$$

The coprime ternary divisor therefore gives

$$2^{JB+1} 3^{v_3(N_p)} \le N_p. \tag{65}$$

Equations (52)–(53) and (60) give

$$N_p = q_p \Omega_p < q_p(\mathcal{K} + 1). \tag{66}$$

Taking binary logarithms in (65)–(66) proves (57). □

Corollary 5 (bilateral exceptional-shell reduction)**.** *Every repeated-mass interval left by the cycle-lap rigidity theorem has a rational plateau anchor on one side of $\mathcal{K}$:*

exceptional interval	*anchor*	*side*
$m = 2$ *exponent-one tower*	$\Xi = (x+1)/q$	$\Xi < \mathcal{K}$,
zero-defect cycle laps	$\Omega = (x+m)/q$	$\Omega \ge \mathcal{K}$.

(67)

The reduced denominator of either anchor is an exact power 3^D which is constant on the interval. Only finitely many exceptional intervals of either kind lie in any bounded depth shell.

For an exceptional interval I, let R_I be its anchor, D_I its depth, and Q_I the maximum of q_j on the interval. If two such intervals have distinct anchors, then

$$\boxed{3^{-\max\{D_I, D_{I'}\}} \le |R_I - R_{I'}| < \frac{1}{Q_I} + \frac{1}{Q_{I'}}.} \tag{68}$$

Hence every infinite family of exceptional repeated-mass intervals must escape through unbounded ternary depth, quantitatively at the scale forced by (40), (56), and (68).

Proof. The two anchor identities and the one-sided inequalities are (34) and (53). The exact denominator and bounded-shell assertions are the preceding two theorems. Tower anchors are strictly increasing. Moreover, on a tower q_j increases at every edge and $\mathcal{K} - \Xi = (e_j - 1)/q_j \le 1/q_j$, so two tower anchors satisfy (68) directly. For two distinct upper anchors, it is (55). For one anchor of each kind, $\Xi < \mathcal{K} \le \Omega$, so

$$\Omega - \Xi = (\mathcal{K} - \Xi) + (\Omega - \mathcal{K}) < \frac{1}{Q_I} + \frac{1}{Q_{I'}}.$$

Their power-of-3 denominators give the lower bound in (68). □

Theorem 6 (universal constant-color run shell)**.** *Retain the normalized fixed-seed setup, and put $c_j := 1 - d_j$. Let $I = [s, t]$ be the state interval of a maximal constant nonpositive-defect run, so that*

$$c_j = c \in \{1, 2, 3\} \qquad (s \le j < t). \tag{69}$$

For $j \in I$, define

$$\begin{aligned} Z_j(c) &:= cx_j + m_j, & \Lambda_I &:= \frac{Z_j(c)}{q_j}, \\ F_j(c) &:= \max\{0, j - v_3(Z_j(c))\}, & F_I &:= F_j(c). \end{aligned} \tag{70}$$

These definitions are independent of the chosen $j \in I$. More precisely,

$$\boxed{2^{a_j} Z_{j+1}(c) = 3Z_j(c), \qquad \frac{Z_{j+1}(c)}{q_{j+1}} = \frac{Z_j(c)}{q_j}, \qquad F_{j+1}(c) = F_j(c).} \tag{71}$$

The reduced denominator of Λ_I is exactly 3^{F_I}.

The anchor lies on a prescribed side of $c\mathcal{K}$:

$$\boxed{\Lambda_I - c\mathcal{K} = \frac{c\theta_j - (c-1)m_j}{q_j},} \tag{72}$$

and hence

c	*side*	*distance bound valid for every* $j \in I$
1	$\Lambda_I \geq \mathcal{K}$	$0 \leq \Lambda_I - \mathcal{K} < 1/q_j,$
2	$\Lambda_I < 2\mathcal{K}$	$0 < 2\mathcal{K} - \Lambda_I \leq m_j/q_j,$
3	$\Lambda_I < 3\mathcal{K}$	$0 < 3\mathcal{K} - \Lambda_I \leq 2m_j/q_j.$

(73)

For each fixed $c \in \{1,2,3\}$ and each integer $F \geq 0$, only finitely many color-c runs have depth $F_I \leq F$.

For a run I, put

$$\varepsilon_I := \begin{cases} \dfrac{1}{\max_{j\in I} q_j}, & c = 1, \\ \min\limits_{j\in I} \dfrac{m_j}{q_j}, & c = 2, \\ 2\min\limits_{j\in I} \dfrac{m_j}{q_j}, & c = 3. \end{cases} \tag{74}$$

If two color-c runs I, I' have distinct anchors, then

$$\boxed{3^{-\max\{F_I, F_{I'}\}} \leq |\Lambda_I - \Lambda_{I'}| < \varepsilon_I + \varepsilon_{I'}.} \tag{75}$$

Moreover, $\varepsilon_I \to 0$ along any sequence of runs whose initial indices tend to infinity. Thus every infinite family of constant nonpositive-defect runs escapes through unbounded exact ternary depth.

Proof. The definition of c_j rewrites the ceiling recurrence as

$$2^{a_j} m_{j+1} = 3m_j - c_j. \tag{76}$$

Multiplying the positive recurrence in (1) by $c_j = c$ and adding (76) proves the first identity in (71). Division by $2^{a_j} q_{j+1} = 3q_j$ proves the second. The first identity also gives

$$v_3(Z_{j+1}(c)) = v_3(Z_j(c)) + 1, \tag{77}$$

which proves the third. Cancellation in (70) shows that the exact reduced denominator is 3^{F_I}. Taking 2-adic valuations in the first identity of (71) and iterating across the run gives

$$\sum_{j=s}^{t-1} a_j = v_2(Z_s(c)) - v_2(Z_t(c)) \leq v_2(Z_s(c)).$$

Since every $a_j \geq 1$, every constant-color run is finite.

Using $e_j = m_j - \theta_j$ and $x_j + e_j = q_j\mathcal{K}$, we obtain

$$\begin{aligned} Z_j(c) &= c(x_j + e_j) + m_j - ce_j \\ &= cq_j\mathcal{K} + c\theta_j - (c-1)m_j, \end{aligned}$$

which proves (72). The bounds in (73) follow from $0 \leq \theta_j < 1$ and $m_j \geq 2$.

We have

$$\frac{m_j}{q_j} \leq \frac{e_j}{q_j} + \frac{1}{q_j} \longrightarrow 0 \tag{78}$$

on the normalized ray. Equations (72)–(73) therefore show that every sequence of late color-c anchors converges to $c\mathcal{K}$. In particular, all color-c anchors lie in one bounded interval.

Fix a depth bound F. Only finitely many rationals in that interval have reduced denominator dividing 3^F. Fix one of them, say R. If $R \neq c\mathcal{K}$, convergence to $c\mathcal{K}$ shows that R occurs only finitely often. If $R = c\mathcal{K}$, then either the equality is impossible because $c\mathcal{K}$ is irrational, or $c\mathcal{K} = u/v$ is rational in lowest terms. In the latter case an occurrence would make

$$q_j c\mathcal{K} = \frac{u3^j}{v2^{A_j}} = Z_j(c) \tag{79}$$

an integer. Its 2-adic valuation tends to $-\infty$ as $A_j \to \infty$, so (79) also occurs only finitely often. This proves bounded-depth finiteness.

For two distinct anchors, their exact power-of-3 denominators give the lower bound in (75). The corresponding line of (73), minimized over each run, gives the upper bound. Finally, (78), together with $q_j \to \infty$ in the color-1 case, proves $\varepsilon_I \to 0$ for late runs. If an infinite family had bounded depths, this would also contradict (75), in agreement with the preceding finiteness argument. □

Corollary 7 (all divergent nonpositive mass escapes every run shell)**.** *On the normalized divergent ray, let $I(j)$ be the maximal constant-color run containing a nonpositive-defect index j. For every integer $F \geq 0$,*

$$\boxed{\sum_{\substack{j \geq 0:\ d_j \leq 0 \\ F_{I(j)} > F}} \frac{1}{m_j} = \infty.} \tag{80}$$

Thus neither simple shadow fragments nor repeated shadow cycles can retain divergent reciprocal mass in a bounded exact ternary shell. The remaining synchronized-shell problem is precisely to exclude divergent mass escaping through $F_I \to \infty$ along one fixed ordinary orbit.

Proof. The total reciprocal mass on $d_j \le 0$ diverges. By the theorem, only finitely many constant-color runs have depth at most F. Every one of those runs is finite, so removing all of them subtracts a finite sum. The remaining sum is (80) and still diverges. □

Theorem 8 (internal switches force maximal run depth)**.** *Let two adjacent maximal nonpositive-defect runs meet at a state index $s \ge 1$, and let their distinct colors be*

$$c, c' \in \{1, 2, 3\}. \tag{81}$$

Let Λ, Λ' and F, F' be their anchors and depths from the universal run-shell theorem, with the primed run entering at s. Then

$$\boxed{\Lambda' - \Lambda = \frac{(c' - c)x_s}{q_s} = \frac{(c' - c)2^{A_s} x_s}{3^s},} \tag{82}$$

and the last fraction has reduced ternary denominator exactly 3^s. More precisely,

$$\boxed{F' = s, \qquad F \le s - 1.} \tag{83}$$

Thus every late internal color switch places the newly entered run in the deepest possible shell at that index. If the new run ends at t, then

$$\boxed{v_3(Z_j(c')) = j - s \qquad (s \le j \le t).} \tag{83a}$$

If it ends at another internal switch from c' to $c'' \ne c'$, put

$$r := t - s, \qquad B := \sum_{j=s}^{t-1} a_j.$$

There is a positive integer H, not divisible by 3, such that

$$\boxed{Z_s(c') = 2^B H, \qquad Z_t(c') = 3^r H.} \tag{83b}$$

Moreover,

$$|c'' - c'| = 1 \Longrightarrow H \text{ is odd}, \qquad |c'' - c'| = 2 \Longrightarrow H \text{ is even}. \tag{83c}$$

Proof. At the common state,

$$Z_s(c') - Z_s(c) = (c' - c)x_s. \tag{84}$$

Division by q_s proves (82). The preceding positive recurrence gives

$$2^{a_{s-1}} x_s = 3x_{s-1} + 1 \equiv 1 \pmod 3, \tag{85}$$

so $3 \nmid x_s$. Also $c' - c \in \{-2, -1, 1, 2\}$, which is a unit modulo 3. The numerator in the last member of (82) is therefore a unit modulo 3, proving the denominator claim.

Let the exiting run begin at $p < s$. Iteration of (77) through this nonempty run gives

$$v_3(Z_s(c)) = v_3(Z_p(c)) + s - p \geq 1.$$

Thus (84), together with the unit property of $(c' - c)x_s$, shows that $Z_s(c')$ is a unit modulo 3. The entering depth is therefore $F' = s$. The exiting depth is already defined at its initial state p, so $F \leq p \leq s - 1$. This proves (83). Finally, iteration of (77) from the unit $Z_s(c')$ proves (83a).

Iteration of the first identity in (71) through the entering run gives

$$2^B Z_t(c') = 3^r Z_s(c').$$

By (83a), $Z_t(c') = 3^r H$ for a positive integer H with $3 \nmid H$. Cancellation gives $Z_s(c') = 2^B H$, which proves (83b). At the exit, $Z_t(c'')$ is even because it begins a color-c'' edge, and

$$Z_t(c'') - Z_t(c') = (c'' - c')x_t.$$

The state x_t is odd. Thus $Z_t(c')$, and hence H, is odd when the color difference is odd and even when it equals 2 in absolute value. This proves (83c). □

Corollary 9 (transition-rich versus entrance-mass dichotomy)**.** *On a normalized divergent ray, exactly one of the following two branches holds.*

1. *There are infinitely many internal switches between colors in* $\{1, 2, 3\}$. *Then infinitely many adjacent run pairs satisfy the maximal-depth law (83).*

2. *There are only finitely many such switches. After deleting a finite prefix, every maximal nonpositive-defect block is one entrance run. The divergent nonpositive reciprocal mass is then entrance-run mass. Hence the run-mass valuation budget forces divergence of the sum of the preceding exponents or of the positive entrance-switch gains.*

*In particular, no contradiction can follow from large run depth or large power-of-*3 *anchor denominator alone: branch* 1 *produces maximal depth automatically. A successful synchronized-shell proof must control reciprocal mass relative to anchor width in branch* 1*, or close the preceding-exponent and switch-gain alternatives in branch* 2*.*

Proof. If the first branch holds, the theorem applies at every internal switch. If it fails, only finitely many nonpositive blocks contain more than their first constant-color run. Removing those finite blocks leaves only entrance runs. Their total mass still diverges, and the stated alternatives are the run-mass valuation budget applied and summed over these runs. The final assertion is the logical content of the two branches. □

Theorem 10 (dyadic numerator law for a ternary run shell)**.** *Retain the normalized fixed-seed setup. Let* $I = [s, t]$ *be the state interval of a constant nonpositive-defect run, with color* $c \in \{1, 2, 3\}$ *and at least one edge. Put*

$$B_I := A_t - A_s, \qquad U_I := v_2(Z_s(c)), \qquad V_I := v_2(Z_t(c)). \tag{86}$$

For $s \le j \le t$*, define*

$$w_j := \min\{j, v_3(Z_j(c))\}, \qquad R_I := \frac{2^{A_j} Z_j(c)}{3^{w_j}}. \tag{87}$$

Then R_I *is a positive integer independent of* j*, and*

$$\boxed{\Lambda_I = \frac{R_I}{3^{F_I}}, \qquad v_2(R_I) = A_j + v_2(Z_j(c)).} \tag{88}$$

In particular,

$$\boxed{A_s + U_I = A_t + V_I, \qquad B_I = U_I - V_I.} \tag{89}$$

Write

$$R_I = 2^{A_t + V_I} \eta_I, \qquad \eta_I \in \mathbb{N} \text{ odd}. \tag{90}$$

The exponent mass of the run has the exact shell decomposition

$$\boxed{B_I = F_I \log_2 3 - A_s + \log_2 \Lambda_I - V_I - \log_2 \eta_I.} \tag{91}$$

If

$$\kappa_1 = \kappa_3 := \frac{1}{3}, \qquad \kappa_2 := \frac{1}{2}, \tag{92}$$

then

$$\boxed{\sum_{j=s}^{t-1} \frac{1}{m_j} \le \kappa_c B_I.} \tag{93}$$

Discard a finite prefix so that $q_s \ge 1$, and put $C := 3\mathcal{K} + 1$. Every remaining run satisfies

$$\boxed{\begin{aligned} B_I &< \log_2(Cq_s) - (s - F_I)\log_2 3, \\ \sum_{j=s}^{t-1} \frac{1}{m_j} &< \kappa_c\big(\log_2(Cq_s) - (s - F_I)\log_2 3\big), \\ F_I &> s - \log_3(Cq_s) \ge s\log_3 2 - \log_3 C. \end{aligned}} \tag{94}$$

Thus linear escape in ternary depth is automatic on every late nonpositive-defect run. The quantity not controlled by depth alone is the positive shell slack in the first line of (94).

Proof. Equation (77) and the identity $A_{j+1} = A_j + a_j$ give

$$w_{j+1} = w_j + 1, \qquad \frac{2^{A_{j+1}} Z_{j+1}(c)}{3^{w_{j+1}}} = \frac{2^{A_j} Z_j(c)}{3^{w_j}}.$$

This proves that R_I is independent of j. Since $F_I = j - w_j$, the anchor identity in (88) follows. Division by a power of 3 does not change a 2-adic valuation, which proves the second identity in (88). Evaluating it at the two endpoints proves (89), and (90) follows.

We can compute the logarithm of R_I in two ways:

$$\begin{aligned} \log_2 R_I &= F_I \log_2 3 + \log_2 \Lambda_I, \\ \log_2 R_I &= A_t + V_I + \log_2 \eta_I = A_s + B_I + V_I + \log_2 \eta_I. \end{aligned}$$

Their equality is exactly (91).

There are at most B_I indices in the run because every exponent is at least 1. On every displayed edge, parity in

$$2^{a_j} m_{j+1} = 3m_j - c$$

shows that m_j is odd for $c \in \{1, 3\}$ and even for $c = 2$. Thus $m_j \ge 3$ in the first case and $m_j \ge 2$ in the second. This proves (93).

The side bounds (73) and $q_s \ge 1$ give

$$0 < \Lambda_I < C.$$

Since $V_I \geq 0$ and $\eta_I \geq 1$, equation (91) now gives

$$0 < B_I < F_I \log_2 3 - A_s + \log_2 C.$$

Using

$$A_s = s \log_2 3 - \log_2 q_s$$

proves the first line of (94), and (93) proves the second. Positivity of B_I gives $F_I > s - \log_3(Cq_s)$. Finally, $A_s \geq s$ implies $q_s \leq (3/2)^s$, and substitution proves the last inequality in (94). □

Corollary 11 (the entrance-exponent alternative is automatic)**.** *Assume the finite-transition branch of the preceding dichotomy, and let $\mathcal{E}$ be the family of all sufficiently late entrance runs. Then*

$$|\mathcal{E}| = \infty, \qquad \sum_{I \in \mathcal{E}} a_{s_I - 1} = \infty. \tag{95}$$

Consequently, divergence of the preceding-exponent sum in the run-mass valuation budget is automatic and cannot distinguish a divergent ray. A necessary condition for the entrance-run mass to diverge is the sharper shell-slack law

$$\boxed{\sum_{I \in \mathcal{E}} \big(\log_2(Cq_{s_I}) - (s_I - F_I)\log_2 3\big) = \infty.} \tag{96}$$

Thus the finite-transition branch has been reduced to controlling this exact shell slack or to a strictly stronger compensated switch-gain bound.

Proof. Every entrance run is finite. Since its reciprocal mass is finite, the divergent total entrance-run mass requires infinitely many distinct entrance runs. Their preceding indices are distinct and every accelerated exponent is at least 1, which proves (95). Summing the second inequality in (94) over $\mathcal{E}$ proves (96), because $\kappa_c \leq 1/2$ for every $c \in \{1, 2, 3\}$. □

Theorem 12 (rational barriers and logarithmic run transport)**.** *Retain the normalized fixed-seed setup, and define*

$$h_j := \frac{x_j}{q_j}, \qquad g_j := \mathcal{K} - h_j = \frac{e_j}{q_j}, \qquad u_j := \frac{m_j}{q_j}. \tag{97}$$

Then

$$\boxed{\begin{aligned} h_{j+1} - h_j &= g_j - g_{j+1} = \frac{1}{3q_j}, \\ u_{j+1} - u_j &= -\frac{c_j}{3q_j}, \\ g_j \leq u_j &< g_j + \frac{1}{q_j} < 2g_j. \end{aligned}} \tag{98}$$

At a fixed state, introduce the three candidate rational barriers

$$\beta_j(c) := h_j + \frac{u_j}{c} = \frac{cx_j + m_j}{cq_j} \qquad (c \in \{1,2,3\}). \tag{99}$$

They satisfy

$$\boxed{\beta_j(3) < \beta_j(2) < \mathcal{K} \le \beta_j(1),} \tag{100}$$

with exact gaps

$$\begin{aligned} \beta_j(1) - \mathcal{K} &= \frac{\theta_j}{q_j}, \\ \mathcal{K} - \beta_j(2) &= \frac{2e_j - m_j}{2q_j}, \\ \mathcal{K} - \beta_j(3) &= \frac{3e_j - m_j}{3q_j}. \end{aligned} \tag{101}$$

Let $I = [s,t]$ be a color-c nonpositive-defect run. Its barrier

$$\beta_I := \frac{\Lambda_I}{c} \tag{102}$$

is constant on the run, and

$$\boxed{u_j = c(\beta_I - h_j), \qquad \frac{u_{j+1}}{u_j} = 1 - \frac{c}{3m_j}.} \tag{103}$$

Consequently,

$$\boxed{\frac{2}{c}\log\frac{\beta_I - h_s}{\beta_I - h_t} \le \sum_{j=s}^{t-1}\frac{1}{m_j} < \frac{3}{c}\log\frac{\beta_I - h_s}{\beta_I - h_t}.} \tag{104}$$

Two useful uniform consequences are

$$\boxed{\begin{aligned} c = 3 \quad &\Longrightarrow \quad \sum_{j=s}^{t-1}\frac{1}{m_j} < \log 2, \\ c = 2 \quad &\Longrightarrow \quad \sum_{j=s}^{r-1}\frac{1}{m_j} < \frac{3}{2}\log 2, \end{aligned}} \tag{105}$$

where in the second line r is the first state of the run with $m_r = 2$, or $r = t$ if no such state occurs. After the first state with $m = 2$, every remaining color-2 edge has exponent 1 and remains at $m = 2$. Thus (105) bounds

the entire color-2 run outside its optional terminal tower. If a color-2 run ends at an internal switch, then the terminal tower is absent, $r = t$, and consequently

$$\boxed{\sum_{j=s}^{t-1} \frac{1}{m_j} < \frac{3}{2} \log 2.} \tag{105a}$$

Finally, suppose that an internal switch at state s changes the color from c to c'. The corresponding barrier jump obeys

$$\boxed{\beta_s(c') - \beta_s(c) = u_s \left(\frac{1}{c'} - \frac{1}{c} \right), \qquad \frac{|\beta_s(c') - \beta_s(c)|}{m_s} = \left| \frac{1}{c'} - \frac{1}{c} \right| \frac{1}{q_s}.} \tag{106}$$

If $\mathcal{S}$ is the set of all internal nonpositive-color switches, then

$$\boxed{\sum_{s \in \mathcal{S}} \frac{|\beta_s(c_s^+) - \beta_s(c_s^-)|}{m_s} \le 2(\mathcal{K} - h_0) < \infty.} \tag{107}$$

Hence the raw weighted barrier-jump charge is already summable. Any remaining transition-rich divergence must be carried by the logarithmic barrier approaches in (104), not by the switch jumps in (106) themselves.

Proof. Division of the positive recurrence by $2^{a_j} q_{j+1} = 3q_j$ gives the first identity in (98). The ceiling recurrence gives the second. Since $m_j = e_j + \theta_j$ and $0 \le \theta_j < 1 < e_j$, the last line of (98) follows.

Equation (99) and $u_j = m_j/q_j$ give the ordering of the three candidate barriers. More precisely, $m_j = e_j + \theta_j$ gives (101). The second and third gaps are positive because $m_j < e_j + 1 < 2e_j < 3e_j$.

On a color-c run, (71) says that Λ_I is constant. Equations (99) and (102) therefore give the first identity in (103). The second line of (98) gives

$$\frac{u_{j+1}}{u_j} = 1 - \frac{c}{3q_j u_j} = 1 - \frac{c}{3m_j},$$

which proves the second identity in (103). Put $z_j := c/(3m_j)$. Parity in the ceiling recurrence gives $0 < z_j \le 1/3$. Hence

$$z_j < -\log(1 - z_j) \le \frac{z_j}{1 - z_j} \le \frac{3}{2} z_j.$$

Sum this inequality and use (103). The result is (104).

Suppose first that $c = 3$, and put $\delta := 3\mathcal{K} - \Lambda_I > 0$. Along the run,

$$u_j = 3g_j - \delta. \tag{108}$$

At the initial state, parity gives $m_s \geq 3$. Therefore

$$\frac{u_s}{g_s} = \frac{m_s}{e_s} < \frac{m_s}{m_s - 1} \leq \frac{3}{2}.$$

Equation (108) now gives $u_s < \delta$. At the terminal state, $u_t \geq g_t$, so (108) gives $g_t \geq \delta/2$ and $u_t \geq \delta/2$. Thus $u_s/u_t < 2$, and the first line of (105) follows from the upper bound in (104).

Now let $c = 2$. If the prefix in (105) is nonempty, then $m_s \geq 4$, so

$$\frac{u_s}{g_s} = \frac{m_s}{e_s} < \frac{m_s}{m_s - 1} \leq \frac{4}{3}.$$

Put $\delta := 2\mathcal{K} - \Lambda_I > 0$. Since $u_j = 2g_j - \delta$, the last inequality gives $u_s < 2\delta$. At the terminal state r of the prefix, $u_r \geq g_r$ gives $u_r \geq \delta$. Therefore $u_s/u_r < 2$, and the second line of (105) follows from (104). If $m_j = 2$ on a color-2 edge, then

$$2^{a_j} m_{j+1} = 4.$$

Since $m_{j+1} \geq 2$ and $a_j \geq 1$, this forces $a_j = 1$ and $m_{j+1} = 2$. This proves the terminal-tower assertion. If the run ends at an internal switch, the next color belongs to $\{1, 3\}$. Reduction of (76) modulo 2 gives $m_t \equiv c_t \pmod 2$, so m_t is odd. A color-2 tower would instead force $m_t = 2$. Hence no such tower occurs and $r = t$, proving (105a).

At an internal switch the two barriers in (99) are evaluated at the same point (h_s, u_s), which proves (106). The largest possible value of $|1/c' - 1/c|$ is $2/3$. Summing the first identity in (98) gives

$$\sum_{j=0}^{\infty} \frac{1}{q_j} = 3(\mathcal{K} - h_0).$$

The second identity in (106), summed over the subset $\mathcal{S}$, now proves (107). □

Theorem 13 (fixed-ray carrier escape and collision law)**.** *Let $I = [s, t]$ be an internally bracketed color-c run on the normalized divergent ray. Put $r := t - s$, and let H be its carrier. Then*

$$\boxed{\frac{H}{m_t} = \frac{1 + cx_t/m_t}{3^r}.} \tag{109}$$

In particular,

$$H \le m_t \quad \Longrightarrow \quad r \log 3 \ge \log\left(1 + \frac{cx_t}{m_t}\right) > \log \frac{x_t}{m_t}. \tag{110}$$

Consequently, for every fixed R, all sufficiently late internally bracketed runs of length at most R satisfy $H > m_t$.

Now let $I' = [t, u]$ be the next run and suppose that it also ends at an internal switch. Write its color as $c' \neq c$, its exponent sum as

$$B' := \sum_{j=t}^{u-1} a_j,$$

and its carrier as H'. At the common state,

$$\boxed{\begin{aligned}(c' - c)x_t &= 2^{B'}H' - 3^r H,\\ (c' - c)m_t &= c'3^r H - c2^{B'}H'.\end{aligned}} \tag{111}$$

If the adjacent carriers collide, $H' = H$, then

$$\boxed{\frac{m_t}{x_t} = \frac{|c'3^r - c2^{B'}|}{|2^{B'} - 3^r|} > \frac{1}{\max\{2^{B'}, 3^r\}},} \tag{112}$$

and hence

$$\max\{B' \log 2, r \log 3\} > \log \frac{x_t}{m_t}. \tag{113}$$

Thus, for every fixed R, only finitely many adjacent carrier collisions can satisfy both $r \le R$ and $B' \le R$. Every infinite collision family on one divergent ray must escape through unbounded run length or unbounded entering exponent sum.

Proof. The terminal identity in (83b) is

$$cx_t + m_t = 3^r H.$$

Division by $3^r m_t$ proves (109), and (110) follows immediately. Equations (32) and (78) give

$$\frac{m_t}{x_t} = \frac{m_t/q_t}{x_t/q_t} \longrightarrow 0,$$

because $x_t/q_t \to \mathcal{K} > 0$. This proves the bounded-length consequence.

At the common state, the terminal identity for I and the initial identity for I' are

$$cx_t + m_t = 3^r H, \qquad c'x_t + m_t = 2^{B'} H'.$$

Subtraction gives the first line of (111). Multiplying the first identity by c', the second by c, and subtracting gives the second line.

Suppose that $H' = H$. The first line of (111) shows that $2^{B'} - 3^r \neq 0$, while the second shows that $c'3^r - c2^{B'} \neq 0$. After division, both factors H and $c' - c$ cancel, which proves the equality in (112). Its numerator is a nonzero integer, so its absolute value is at least 1. Its denominator is strictly smaller than $\max\{2^{B'}, 3^r\}$. This proves the strict inequality in (112) and, after taking logarithms, (113). The limit $m_t/x_t \to 0$ proves the final finiteness assertion. □

Theorem 14 (finite strict-barrier carrier counterexamples)**.** *Neither a terminal bound $H \geq m_t$ nor injectivity of the carriers of adjacent internally bracketed runs follows from the synchronized recurrences, the ceiling condition, and the strict-barrier inequalities.*

More precisely, the finite data

j	0	1	2	3
x_j	91	137	103	155
m_j	48	71	53	78
θ_j	26/27	17/18	17/24	1/16

$$(a_0, a_1, a_2) = (1, 2, 1)$$

give a synchronized real companion with color word $(2, 1, 3)$. The middle color-1 run is internally bracketed and has carrier

$$x_1 + m_1 = 208 = 2^2 \cdot 52, \qquad x_2 + m_2 = 156 = 3 \cdot 52.$$

Thus $H = 52 < m_2 = 53$.

There is also a strict-barrier synchronized segment

j	0	1	2	3	4	5
x_j	27	41	31	47	71	107
m_j	29	43	32	47	70	104
θ_j	1/2	3/4	9/16	11/32	33/64	35/128

with exponent word $(1, 2, 1, 1, 1)$ and color word $(1, 1, 2, 1, 2)$. Its two adjacent internally bracketed runs have colors 2 and 1, and both have carrier 47:

$$2x_2 + m_2 = 94 = 2 \cdot 47, \quad 2x_3 + m_3 = 141 = 3 \cdot 47,$$
$$x_3 + m_3 = 94 = 2 \cdot 47, \quad x_4 + m_4 = 141 = 3 \cdot 47.$$

Hence any carrier charging argument must use fixed-ray information beyond finite strict-barrier synchronization.

Proof. For the first segment, direct calculation gives

$$2^{a_j} x_{j+1} = 3x_j + 1, \qquad 2^{a_j} m_{j+1} = 3m_j - c_j$$

with $(c_0, c_1, c_2) = (2, 1, 3)$. The displayed rounding errors satisfy

$$2^{a_j} \theta_{j+1} = 3\theta_j + 1 - c_j,$$

and lie in $[0, 1)$. Therefore $e_j := m_j - \theta_j > 1$ and subtraction proves $2^{a_j} e_{j+1} = 3e_j - 1$. The cumulative exponents are $(1, 3, 4)$, so

$$2^1 < 3, \qquad 2^3 < 3^2, \qquad 2^4 < 3^3.$$

The carrier calculation in the statement proves the first assertion.

For the second segment, the same three direct identities hold with the displayed exponent, color, and rounding-error words. Its cumulative exponents are $(1, 3, 4, 5, 6)$, and

$$2, 8, 16, 32, 64 < 3, 9, 27, 81, 243$$

coordinatewise. Thus this word is also a strict barrier. The two carrier calculations prove the failure of local injectivity. □

Theorem 15 (projective defect charge and zero-defect reduction)**.** *Retain the normalized fixed-seed setup, and define the projective rounding coordinate*

$$p_j := \frac{\theta_j}{m_j}. \tag{114}$$

On every nonpositive-defect edge, with $c_j = 1 - d_j \in \{1, 2, 3\}$, we have

$$\boxed{p_{j+1} - p_j = \frac{(1 - c_j) m_j + c_j \theta_j}{m_j (3m_j - c_j)}.} \tag{115}$$

Consequently,

edge type	*projective charge*
$c_j = 1$	$0 \le p_{j+1} - p_j < \dfrac{1}{8m_j},$
$c_j = 2,\ m_j \ge 4$	$\dfrac{1}{m_j} < 5(p_j - p_{j+1}),$
$c_j = 3$	$\dfrac{1}{m_j} < 2(p_j - p_{j+1}).$

(116)

Let $\mathcal{B}$ be the family of maximal consecutive blocks of nonpositive-defect edges. For $B = [\alpha, \beta) \in \mathcal{B}$, put

$$\begin{aligned} \mathcal{Z}_B &:= \{j \in B : c_j = 1\}, \\ \mathcal{T}_B &:= \{j \in B : c_j = 2,\ m_j = 2\}, \\ \mathcal{R}_B &:= B \setminus (\mathcal{Z}_B \cup \mathcal{T}_B). \end{aligned} \tag{117}$$

Then

$$\boxed{\sum_{j \in \mathcal{R}_B} \frac{1}{m_j} \le \frac{5}{m_\alpha} + \frac{5}{8} \sum_{j \in \mathcal{Z}_B} \frac{1}{m_j}.} \tag{118}$$

The same inequality holds for a final infinite block, with the sums interpreted as monotone limits.

Put

$$\mathcal{E} := \{\alpha : [\alpha, \beta) \in \mathcal{B}\}, \qquad \mathcal{Z} := \bigcup_{B \in \mathcal{B}} \mathcal{Z}_B, \qquad \mathcal{T} := \bigcup_{B \in \mathcal{B}} \mathcal{T}_B. \tag{119}$$

Since the total nonpositive-defect reciprocal mass diverges, at least one of the following three series diverges:

$$\boxed{\sum_{j \in \mathcal{E}} \frac{1}{m_j}, \qquad \sum_{j \in \mathcal{Z}} \frac{1}{m_j}, \qquad \sum_{j \in \mathcal{T}} \frac{1}{m_j}.} \tag{120}$$

Finally, let $I = [s, t]$ be a color-1 run with $\theta_s > 0$. Then

$$\boxed{\ell_I := \log \frac{p_t}{p_s} = \sum_{j=s}^{t-1} -\log\left(1 - \frac{1}{3m_j}\right), \qquad 2\ell_I \le \sum_{j=s}^{t-1} \frac{1}{m_j} < 3\ell_I.} \tag{121}$$

Only finitely many color-1 runs on the normalized ray can have $\theta_s = 0$. Therefore the middle alternative in (120) is equivalent to

$$\sum_{I \text{ a color-1 run}} \ell_I = \infty. \tag{122}$$

More generally, define

$$P_1(p) := p, \qquad P_2(p) := 1 - 2p, \qquad P_3(p) := 2 - 3p. \tag{123}$$

These quantities are nonnegative on an edge of the corresponding color. Whenever $P_c(p_j) > 0$, we have

$$\boxed{\frac{P_c(p_{j+1})}{P_c(p_j)} = \left(1 - \frac{c}{3m_j}\right)^{-1}.} \tag{124}$$

Hence, for every color-c run $I = [s, t]$ with $P_c(p_s) > 0$,

$$L_I := \log \frac{P_c(p_t)}{P_c(p_s)} \qquad \Longrightarrow \qquad \frac{2}{c} L_I \le \sum_{j=s}^{t-1} \frac{1}{m_j} < \frac{3}{c} L_I. \tag{125}$$

In particular, an $m = 2$ tower of K edges satisfies

$$\log \frac{P_2(p_{s+K})}{P_2(p_s)} = K \log \frac{3}{2}, \qquad \sum_{j=s}^{s+K-1} \frac{1}{m_j} = \frac{K}{2}. \tag{126}$$

Thus all divergent nonpositive mass is forced into block entrances, logarithmic amplification of P_1 on zero-defect runs, or logarithmic amplification of P_2 on terminal towers.

Proof. The ceiling and rounding recurrences are

$$2^{a_j} m_{j+1} = 3m_j - c_j, \qquad 2^{a_j} \theta_{j+1} = 3\theta_j + 1 - c_j.$$

We divide the second identity by the first and subtract $p_j = \theta_j / m_j$. This proves (115).

If $c_j = 1$, equation (115) gives

$$p_{j+1} - p_j = \frac{\theta_j}{m_j(3m_j - 1)}.$$

Parity gives $m_j \ge 3$, and $0 \le \theta_j < 1$, which proves the first line of (116). If $c_j = 2$, then

$$p_j - p_{j+1} = \frac{m_j - 2\theta_j}{m_j(3m_j - 2)}.$$

For $m_j \ge 4$, we have

$$\frac{1/m_j}{p_j - p_{j+1}} = \frac{3m_j - 2}{m_j - 2\theta_j} < \frac{3m_j - 2}{m_j - 2} \le 5.$$

The function in the last member is decreasing for $m_j \ge 4$. Finally, if $c_j = 3$, parity gives $m_j \ge 3$, and

$$\frac{1/m_j}{p_j - p_{j+1}} = \frac{3m_j - 3}{2m_j - 3\theta_j} < \frac{3m_j - 3}{2m_j - 3} \le 2.$$

This proves (116).

Fix a block $B = [\alpha, \beta)$. Let U_B be the total increase of p_j on $\mathcal{Z}_B$, and let D_B and D_B^{tower} be its total decreases on $\mathcal{R}_B$ and $\mathcal{T}_B$, respectively. Telescoping over a finite initial part of the block gives

$$0 \le p_{\text{end}} = p_\alpha + U_B - D_B - D_B^{\text{tower}}.$$

Therefore $D_B \le p_\alpha + U_B$. We have

$$p_\alpha < \frac{1}{m_\alpha}, \qquad U_B < \frac{1}{8} \sum_{j \in \mathcal{Z}_B} \frac{1}{m_j}.$$

The last two lines of (116), with the weaker common constant 5, now prove (118). Passing through finite initial parts proves the infinite-block case. Summing (118) over $\mathcal{B}$ shows that convergence of all three series in (120) would force convergence on $\bigcup_B \mathcal{R}_B$ as well. This would contradict the divergent nonpositive-defect mass. Hence (120) holds.

On a color-1 edge with $p_j > 0$, direct division gives

$$\frac{p_{j+1}}{p_j} = \frac{3m_j}{3m_j - 1} = \left(1 - \frac{1}{3m_j}\right)^{-1}.$$

Multiplication along the run proves the identity in (121). We also have

$$0 < \frac{1}{3m_j} \le \frac{1}{9}, \qquad \frac{1}{3m_j} < -\log\left(1 - \frac{1}{3m_j}\right) \le \frac{1}{2m_j}.$$

This proves the two-sided estimate. If $\theta_s = 0$, every color-1 edge preserves $\theta = 0$, and the corresponding run anchor equals $\mathcal{K}$. The argument surrounding (79) proves that this equality can occur only finitely many times on the normalized ray. Removing those finitely many runs and summing (121) proves (122).

It remains to prove the unified identity. For $c = 1$, equation (124) is the ratio already used in (121). For $c = 2$, we have

$$2^{a_j}(m_{j+1} - 2\theta_{j+1}) = 3(m_j - 2\theta_j),$$

while for $c = 3$ we have

$$2^{a_j}(2m_{j+1} - 3\theta_{j+1}) = 3(2m_j - 3\theta_j).$$

Division by the corresponding ceiling recurrence proves (124). We sum its logarithm along a run and apply the inequality used in (104), which proves (125). On an $m = 2$ tower, every term in (124) equals $3/2$, and every reciprocal term equals $1/2$. This proves (126). □

Theorem 16 (threshold-reset identity for internal zero-defect runs)**.** *Retain the normalized fixed-seed setup. Let $I = [s, t]$ be a maximal color-1 run with $\theta_s > 0$. Suppose that it is internally bracketed by nonpositive-defect edges of colors*

$$c_- := c_{s-1} \in \{2, 3\}, \qquad c_+ := c_t \in \{2, 3\}. \tag{127}$$

We define its entrance threshold residual and size-corrected reset depth by

$$\begin{aligned} \rho_I &:= 3\theta_{s-1} - (c_- - 1) = 2^{a_{s-1}}\theta_s, \\ \Gamma_I &:= \log \frac{3m_{s-1} - c_-}{m_t \rho_I}. \end{aligned} \tag{128}$$

Then $\rho_I > 0$ and

$$\boxed{p_s = \frac{\rho_I}{3m_{s-1} - c_-}, \qquad \Gamma_I = \log \frac{m_s}{m_t \theta_s}.} \tag{129}$$

The exact projective amplification from (121) satisfies

$$\boxed{\ell_I = \Gamma_I + \log \theta_t, \qquad 0 < \Gamma_I - \ell_I \le \log \frac{3}{c_+ - 1}.} \tag{130}$$

Equality holds in the upper bound precisely when the edge following the run has $\theta_{t+1} = 0$.

Consequently, for every collection $\mathcal{I}$ of internally bracketed color-1 runs,

$$\sum_{I \in \mathcal{I}} \sum_{j=s(I)}^{t(I)-1} \frac{1}{m_j} = \infty \quad \Longrightarrow \quad \sum_{I \in \mathcal{I}} \ell_I = \infty \quad \Longrightarrow \quad \sum_{I \in \mathcal{I}} \Gamma_I = \infty. \tag{131}$$

Thus any summable upper charge for the reset depths Γ_I closes the internally bracketed part of the zero-defect branch.

Proof. On the edge entering the run, the ceiling and rounding recurrences give

$$2^{a_{s-1}} m_s = 3m_{s-1} - c_-, \qquad 2^{a_{s-1}} \theta_s = 3\theta_{s-1} + 1 - c_- = \rho_I. \tag{132}$$

The assumption $\theta_s > 0$ proves $\rho_I > 0$. Dividing the two identities in (132) proves the first identity in (129), and substituting them in (128) proves the second.

At the endpoints of the color-1 run, we have

$$\begin{aligned} \ell_I &= \log \frac{p_t}{p_s} = \log \frac{m_s \theta_t}{m_t \theta_s} \\ &= \Gamma_I + \log \theta_t. \end{aligned} \tag{133}$$

The edge following the run has color c_+, so

$$2^{a_t}\theta_{t+1} = 3\theta_t + 1 - c_+ \geq 0. \tag{134}$$

Hence

$$\frac{c_+ - 1}{3} \leq \theta_t < 1. \tag{135}$$

Equations (133)–(135) prove (130), including its equality statement. Finally, (121) compares the reciprocal mass of each run with ℓ_I, while (130) gives $\Gamma_I > \ell_I$. Summation proves (131). □

Theorem 17 (strict-barrier failure of local reset bounds)**.** *For every integer $K \geq 1$, the exponent word*

$$w_K := \big(1, (1,2)^K, 1\big) \tag{136}$$

has a positive odd ordinary realization and a synchronized real companion segment with ceiling and color data

$$\begin{aligned} (m_0, m_1, \ldots, m_{2K+2}) &= \big(4, 5, (7,5)^K, 6\big), \\ (c_0, c_1, \ldots, c_{2K+1}) &= \big(2, 1^{2K}, 3\big). \end{aligned} \tag{137}$$

Its middle $2K$ edges form one internally bracketed color-1 run $I_K = [1, 2K+1]$. This run consists of K laps of

$$5 \xrightarrow{a=1} 7 \xrightarrow{a=2} 5 \tag{138}$$

and satisfies

$$\boxed{\begin{aligned} \sum_{j=1}^{2K} \frac{1}{m_j} &= \frac{12K}{35}, \\ \ell_{I_K} &= K \log \frac{9}{8}, \\ \rho_{I_K} &= \frac{4}{3}\left(\frac{8}{9}\right)^K, \\ \Gamma_{I_K} &= K \log \frac{9}{8} + \log \frac{3}{2}. \end{aligned}} \tag{139}$$

Every prefix of w_K obeys the strict-barrier inequality

$$2^{A_n} < 3^n, \qquad A_n := \sum_{j<n} a_j. \tag{140}$$

In particular, the complete discrete boundary tuple

$$(c_-, a_{s-1}, m_{s-1}, m_s, m_t, c_+, a_t, m_{t+1}) = (2, 1, 4, 5, 5, 3, 1, 6) \tag{141}$$

is independent of K, while both the run mass and its projective amplification tend to infinity. Therefore no finite-valued bound depending only on this boundary tuple, or on any fixed-width window of exponent, color, and ceiling data at the two boundaries, can control zero-defect mass or projective amplification on finite strict-barrier synchronized segments. A valid bound must detect the threshold residual in (128), or use coherence along one fixed ordinary ray across different runs.

Proof. We first construct the real companion. Set

$$\theta_1 := \frac{2}{3}\left(\frac{8}{9}\right)^K, \qquad \theta_0 := \frac{1 + 2\theta_1}{3}. \tag{142}$$

Use the ceiling path in (137). The first edge satisfies

$$2 \cdot 5 = 3 \cdot 4 - 2, \qquad 2\theta_1 = 3\theta_0 - 1. \tag{143}$$

On each color-1 lap in (138), the ceiling identities are

$$2 \cdot 7 = 3 \cdot 5 - 1, \qquad 4 \cdot 5 = 3 \cdot 7 - 1, \tag{144}$$

and the corresponding two rounding updates multiply θ by $(3/2)(3/4) = 9/8$. Hence

$$\theta_{2K+1} = \left(\frac{9}{8}\right)^K \theta_1 = \frac{2}{3}. \tag{145}$$

The final color-3 edge satisfies

$$2 \cdot 6 = 3 \cdot 5 - 3, \qquad 2\theta_{2K+2} = 3\theta_{2K+1} - 2 = 0. \tag{146}$$

All rounding errors lie in $[0, 1)$. Indeed, before the second half of the last lap the error is $16/27$, after its first half it is $8/9$, and all earlier errors are smaller. Also $1/3 < \theta_0 < 1$. Thus $e_j := m_j - \theta_j > 1$, and (143)–(146) prove the synchronized real recurrence.

We next verify the strict barrier. Put $\lambda := \log_2 3$. After the initial exponent and k complete pairs, the barrier surplus is

$$\lambda(1 + 2k) - (1 + 3k) = \lambda - 1 + (2\lambda - 3)k > 0. \tag{147}$$

After the first exponent of the next pair, and also after the final extra exponent, it is

$$\lambda(2+2k)-(2+3k)=2(\lambda-1)+(2\lambda-3)k>0. \tag{148}$$

Here $\lambda>1$ follows from $3>2$, and $2\lambda>3$ follows from $3^2>2^3$. Equations (147)–(148) cover every prefix and prove (140).

For completeness, we prove that every finite positive exponent word $(a_0,\dots,a_{N-1})$ has a positive odd ordinary realization. Put

$$A_k:=\sum_{j<k}a_j, \qquad C_k:=\sum_{i=0}^{k-1}3^{k-1-i}2^{A_i} \qquad (0\le k\le N). \tag{149}$$

Iteration of the ordinary recurrence gives

$$2^{A_k}x_k=3^kx_0+C_k \qquad (0\le k\le N). \tag{150}$$

Since 3^N is invertible modulo 2^{A_N+1}, there is a unique residue class x_0 modulo 2^{A_N+1} satisfying

$$3^Nx_0+C_N\equiv 2^{A_N} \pmod{2^{A_N+1}}. \tag{151}$$

The integer C_N is odd, so this residue class is odd. We take its least positive representative.

We check all intermediate valuations. For $k<N$, split C_N at k and reduce (151) modulo 2^{A_k+1}. The term with index $i=k$ is the only tail term not already divisible by 2^{A_k+1}, and we obtain

$$3^{N-1-k}\left(3(3^kx_0+C_k)+2^{A_k}\right)\equiv 0 \pmod{2^{A_k+1}}. \tag{152}$$

Multiplication by odd units does not change the unique nonzero class 2^{A_k} modulo 2^{A_k+1}. Therefore

$$3^kx_0+C_k\equiv 2^{A_k} \pmod{2^{A_k+1}}. \tag{153}$$

Together with (151), this proves that every quotient $x_k:=(3^kx_0+C_k)/2^{A_k}$ is a positive odd integer. Consecutive instances of (150) give

$$2^{a_k}x_{k+1}=3x_k+1,$$

with the prescribed exact exponent. Applying this construction to w_K proves the ordinary realization asserted in the theorem.

It remains to calculate the charges. Each lap contributes $1/5 + 1/7 = 12/35$ to the reciprocal mass. Its two rounding updates multiply p by $9/8$, because the initial and terminal ceiling of a lap are both 5. This proves the first two identities in (139). Equations (142)–(143) give

$$\rho_{I_K} = 3\theta_0 - 1 = 2\theta_1 = \frac{4}{3}\left(\frac{8}{9}\right)^K. \tag{154}$$

Finally, $m_s = m_t = 5$ and (129) give

$$\Gamma_{I_K} = \log\frac{1}{\theta_1} = K\log\frac{9}{8} + \log\frac{3}{2}. \tag{155}$$

This proves (139). The tuple (141) is read directly from (136)–(137). For any fixed boundary-window width, taking K larger than that width leaves the two discrete boundary windows unchanged while (139) grows without bound. The claimed impossibility follows. □

Theorem 18 (global logarithmic separation clock)**.** *Retain the normalized fixed-seed setup and define*

$$\mathcal{L}_j := \log\frac{q_j}{m_j}. \tag{156}$$

For every edge, with $c_j := 1 - d_j$, one has

$$\boxed{\mathcal{L}_{j+1} - \mathcal{L}_j = \log\frac{3m_j}{3m_j - c_j} = -\log\left(1 + \frac{d_j - 1}{3m_j}\right).} \tag{157}$$

The sign of this increment is determined exactly by the defect:

d_j	c_j	$\mathcal{L}_{j+1} - \mathcal{L}_j$
$d_j \le 0$	$c_j \in \{1,2,3\}$	> 0
$d_j = 1$	$c_j = 0$	$= 0$
$d_j \ge 2$	$c_j \le -1$	$< 0.$

(158)

On a nonpositive-defect edge,

$$\boxed{\frac{2}{c_j}(\mathcal{L}_{j+1} - \mathcal{L}_j) \le \frac{1}{m_j} < \frac{3}{c_j}(\mathcal{L}_{j+1} - \mathcal{L}_j).} \tag{159}$$

Consequently, every consecutive nonpositive-defect block $B = [\alpha, \beta)$ satisfies

$$\boxed{\frac{2}{3}(\mathcal{L}_\beta - \mathcal{L}_\alpha) \le \sum_{j=\alpha}^{\beta-1}\frac{1}{m_j} < 3(\mathcal{L}_\beta - \mathcal{L}_\alpha).} \tag{160}$$

The same assertion holds for a final infinite block, using monotone limits.

Let $I = [s,t]$ be a constant nonpositive-defect run of color $c \in \{1,2,3\}$, and let Λ_I be its rational anchor from (70). Put

$$\Delta_I := \begin{cases} \Lambda_I - \mathcal{K}, & c = 1, \\ c\mathcal{K} - \Lambda_I, & c \in \{2,3\}. \end{cases} \tag{161}$$

Whenever $\Delta_I > 0$, the projective potential in (123) has the exact factorization

$$\boxed{P_c(p_j) = \Delta_I \frac{q_j}{m_j} = \Delta_I e^{\mathcal{L}_j} \qquad (s \le j \le t).} \tag{162}$$

In particular, the projective run gain in (125) is

$$\boxed{L_I = \mathcal{L}_t - \mathcal{L}_s.} \tag{163}$$

For an internally bracketed color-1 run with positive entrance rounding, the quantities in the threshold-reset theorem therefore satisfy

$$\boxed{\ell_I = \mathcal{L}_t - \mathcal{L}_s, \qquad \Gamma_I = \mathcal{L}_t - \mathcal{L}_s - \log \theta_t.} \tag{164}$$

Define the accumulated nonpositive gain and positive-defect contraction by

$$\begin{aligned} \mathcal{C}_N &:= \sum_{\substack{0 \le j < N \\ d_j \le 0}} -\log\left(1 + \frac{d_j - 1}{3m_j}\right), \\ \mathcal{P}_N &:= \sum_{\substack{0 \le j < N \\ d_j \ge 2}} \log\left(1 + \frac{d_j - 1}{3m_j}\right). \end{aligned} \tag{165}$$

Then

$$\boxed{\mathcal{L}_N - \mathcal{L}_0 = \mathcal{C}_N - \mathcal{P}_N,} \tag{166}$$

and

$$\frac{2}{3}\mathcal{C}_N \le \sum_{\substack{0 \le j < N \\ d_j \le 0}} \frac{1}{m_j} \le 3\mathcal{C}_N. \tag{167}$$

Moreover,

$$\boxed{\begin{aligned} &\log \frac{x_{j+1}/m_{j+1}}{x_j/m_j} \\ &\qquad = \mathcal{L}_{j+1} - \mathcal{L}_j + \log\left(1 + \frac{1}{3x_j}\right). \end{aligned}} \tag{168}$$

Since $\sum_j 1/x_j < \infty$, the last terms in (168) have a finite total. Thus

$$\boxed{\sum_{d_j \le 0} \frac{1}{m_j} = \infty \quad \Longleftrightarrow \quad \sum_{d_j \le 0} \log \frac{x_{j+1}/m_{j+1}}{x_j/m_j} = \infty.} \tag{169}$$

Finally, $m_j/q_j \to 0$, so $\mathcal{L}_j \to \infty$. Hence every normalized divergent ray obeys the strict drift law

$$\boxed{\mathcal{C}_N - \mathcal{P}_N \longrightarrow \infty.} \tag{170}$$

Therefore all divergent nonpositive reciprocal mass, including zero-defect cycle laps and terminal $m = 2$ towers, is quantitatively equivalent to the positive variation of the single clock $\mathcal{L}_j$. A fixed-ray closure must contradict (170), not merely bound one color in isolation.

Proof. The ceiling recurrence is

$$2^{a_j} m_{j+1} = 3m_j - c_j,$$

while $q_{j+1}/q_j = 3/2^{a_j}$. Division proves (157), and the three signs in (158) follow immediately.

Suppose that $d_j \le 0$. Then $c_j \in \{1, 2, 3\}$. Put

$$z := \frac{c_j}{3m_j}.$$

Parity in the ceiling recurrence gives $m_j \ge c_j$, and hence $0 < z \le 1/3$. The elementary estimates

$$z < -\log(1 - z) \le \frac{z}{1 - z} \le \frac{3}{2} z$$

now prove (159). Summing over a block and using $1 \le c_j \le 3$ proves (160).

Equation (72) gives

$$\Lambda_I - c\mathcal{K} = \frac{c\theta_j - (c - 1)m_j}{q_j}.$$

For $c = 1$, this says $P_1(p_j) = \theta_j/m_j = (\Lambda_I - \mathcal{K})q_j/m_j$. For $c \in \{2, 3\}$, it says

$$P_c(p_j) = (c - 1) - cp_j = (c\mathcal{K} - \Lambda_I)\frac{q_j}{m_j}.$$

This proves (162). Taking endpoint ratios proves (163), and (164) follows from (130).

Equations (166) and (167) are obtained by summing (157) and (159), noting that a defect $d_j = 1$ contributes zero. Next, $h_j = x_j/q_j$ and (98) give

$$\frac{h_{j+1}}{h_j} = 1 + \frac{1}{3x_j}.$$

Since

$$\log \frac{x_j}{m_j} = \mathcal{L}_j + \log h_j,$$

equation (168) follows. The inequality $\log(1+u) \le u$ and the convergence of $\sum_j 1/x_j$ show that the total correction in (168) is finite. On every non-positive edge, both terms on the right of (168) are positive. Combining this observation with (167) proves (169).

Finally, equation (78) gives $m_j/q_j \to 0$. Therefore $\mathcal{L}_j = \log(q_j/m_j) \to \infty$, and (166) proves (170). □

Corollary 19 (upper-anchor defect-tail identity)**.** *For every state index j, the ordered defect tail converges and satisfies*

$$\boxed{\frac{\theta_j}{q_j} = \Omega_j - \mathcal{K} = -\sum_{k=j}^{\infty} \frac{d_k}{3q_k}.} \tag{171}$$

Let $I = [s,t]$ be an internally bracketed color-1 run as in the threshold-reset theorem. Then

$$\boxed{\rho_I = q_{s-1}\left(\frac{c_+ - 1}{q_t} - \sum_{k=t+1}^{\infty} \frac{d_k}{q_k}\right).} \tag{172}$$

Equivalently,

$$\boxed{\rho_I = q_{s-1}\left(\frac{c_+ - 1}{q_t} + \frac{3\theta_{t+1}}{q_{t+1}}\right).} \tag{173}$$

Thus the entrance residual is exactly the exit threshold jump, transported back to the entering scale, together with the complete future defect tail.

Proof. Equation (53) gives

$$\Omega_{k+1} - \Omega_k = \frac{d_k}{3q_k}.$$

Also $\Omega_k - \mathcal{K} = \theta_k/q_k \to 0$. Telescoping from j to N and letting $N \to \infty$ proves (171), without any assertion of absolute convergence.

On the run, $d_k = 0$ for $s \le k < t$, while $d_t = 1 - c_+ = -(c_+ - 1)$. Substitution in (171), followed by $\rho_I = 3q_{s-1}(\Omega_s - \mathcal{K})$ from (128)–(129), proves (172). Applying (171) at $t+1$ gives

$$\sum_{k=t+1}^{\infty} \frac{d_k}{q_k} = -\frac{3\theta_{t+1}}{q_{t+1}},$$

which proves (173). □